

operating systems based on the UNIX code. The Mac OS, and early versions of Windows too, were based on the UNIX code, the edge being in the GUI they provided.

BSD was one of the first operating systems to embrace the concept of open source and free software. BSD's license is still preferred on some points over the GPL license used by Linux, because it allows third party propriety software to interface with the operating system. BSD allows for derivative works to be used for commercial purposes, something specifically disallowed by the GPL, and thus many consider the BSD license to be more free than the GPL. BSD was the first operating system to implement IP protocols, which enabled a smoother connection between computers, and is the standard now, as this allowed files to be written and accessed across a network as smoothly as a native hard drive.

GNU (Gnu's Not Linux) and Minix were two important Unix clones. GNU pioneered the free software movement, making sure that the operating system itself, and all derivative works were free and open source for everyone. This meant that no one could ever use GNU for commercial purposes. The project lapsed into technical difficulties and was never successful, except in the spread of its philosophy. Minix was an operating system developed for educational purposes. Andrew S Tanenbaum released the operating system under the BSD license, to accompany his text book on writing operating systems. Minix gathered a lot of interest, and was one of the inspirations for Linux. Note that both GNU and Minix used no code from UNIX, and were totally different operating systems, although they are referred to as being UNIX-like.

In 1991, Linus Trovalds used the Minix usenet group to introduce, discuss and reveal the early versions of the Linux Kernel. Linus began to use Minix on his home computer, because he wanted to use UNIX, on which he was working at the University of Helsinki. But the source code of Minix was not entirely available, so Linus started writing his own. What followed was one of the earliest and most infamous flamewars in Internet history, between Trovalds and Tanenbaum. The issue was a very technical one, on what kind of operating system was better, a monolithic kernel like Linux, or a microkernel based OS like Minix. A microkernel excludes elements like device drivers and inter-process communication from the scope of the operating system, putting the onus on the user. This debate has still not settled, but Linux is by far the more proven and popular operating system. To Tanenbaum's cred-